Forehand joint instability arises from fatigue-induced motor control loss

Nathan To

Department of Mechanical Engineering, National Taiwan University, Taipei, Taiwan

Abstract:

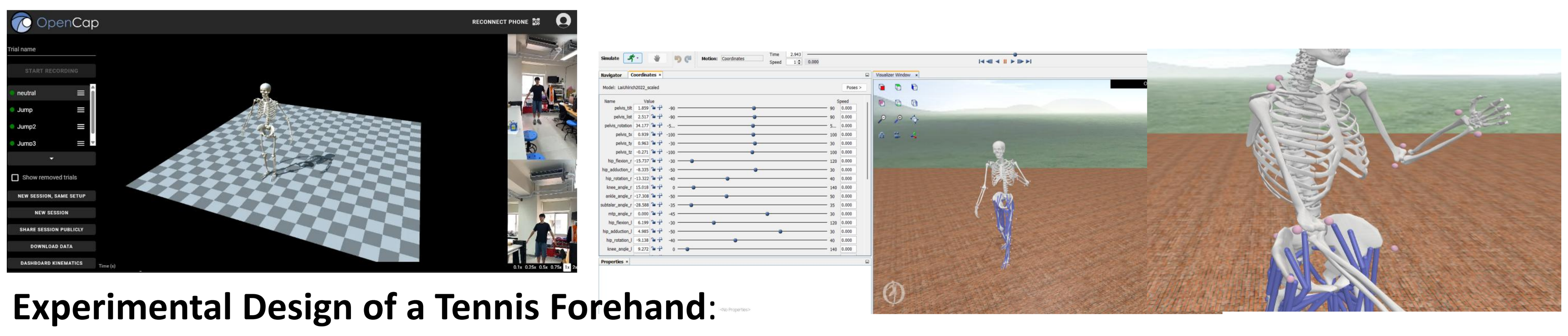
Repetitive upper limb motions such as the tennis “forehand” swing are highly subject to neuromuscular fatigue, which can disrupt coordinated joint control movement and increase the risk of overuse injuries. Despite the significance of elbow and shoulder collaboration movement within dynamic sports, the effects of fatigue on those joints persists in complex movement. In my study, I examined the impact of fatigue on the upper right limb kinematics during a tennis forehand swing using motion data collecting software; OpenSim and OpenCap. Normalizing joint angles and their trajectories from the shoulder (flexion, adduction, rotation) and elbow (flexion) were compared across a baseline found using several methods of finding the Euclidean Distance and most importantly, dynamic time warping. My analysis revealed increase inconsistency and reduced peak joint angles in the “fatigue” state, rather distinctly in elbow flexion and shoulder adduction. Examining the fatigue trials, it exhibited greater volatility across all repetitions, suggesting compromised motor control and compensatory joint behavior. Concluding, these fatigue movements mediate disruption in joint coordination during the forehand motion which may contribute to a decreased performance and a higher injury risk. Understanding the caution and breaking down the movement of motions can intervene strategies in optimization for robotic movements and most importantly, athletic training for injury prevention.

Keywords: Fatigue, Forehand motion, joint kinematics, elbow flexion, motor control, neuromuscular fatigue

Background

- **Execution**:
  - A tennis forehand is executed by rotating the trunk and hips during the backswing, initiating a chain that transfers force through the shoulder, elbow, and wrist as the racquet contacts the ball in front of the body. Once contact hits, it is then followed by a controlled follow-through to stabilize joint motion.
- **Analyze** the biomechanics of a forehand swing to understand joint behavior under stress, fatigue, and rest using motion capture software to predict outcomes like injury risk (Uhlrich et al. 2023).

Methods



Experimental Design of a Tennis Forehand:

- Trial consisted of a series of swings categorizing baseline trials, and fatigue trials where this data was then exported into a .csv file to measure joint angle measurements across time.
- Final graphs had to undergo data processing to “smooth” out the lines for more evident data interpretation.

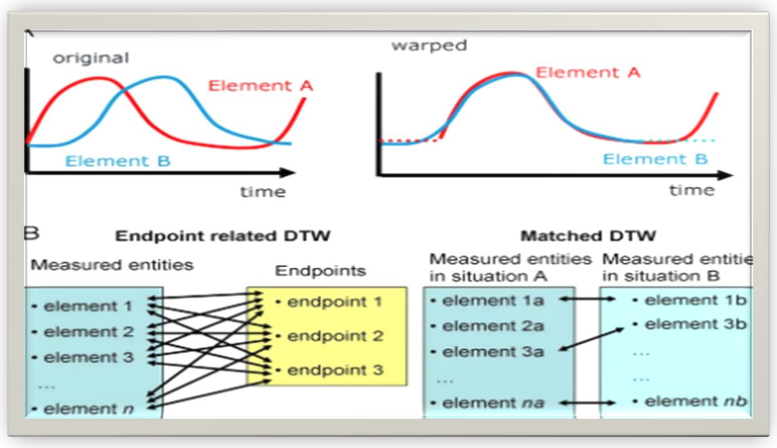
- **Baseline vs. Fatigue:**
  - A **baseline** refers to an average measured joint motion during a swing under well rest.
  - A **fatigue trial** underwent exertion through repetitive motion playtime beforehand, running for around 30 min.
- **Data Collection:**
  - Motion capture data (.csv files) were collected using OpenCap during multiple trials.
    - arm_flex_r (Shoulder flexion)
    - arm_add_r (Shoulder adduction)
    - arm_rot_r (Shoulder rotation)
    - elbow_flex_r (Elbow Flexion)

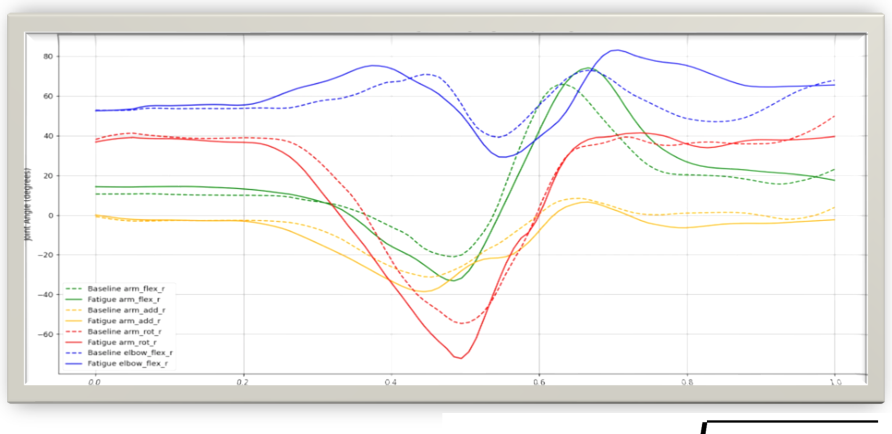
Formula: $t_{norm} = \frac{t - t_{\min}}{t_{\max} - t_{\min}}$

Equipment:

- A neutral model of a real-time scan of my body was made to simulate the forehand swing captured by 3 iOS cameras linked up together using OpenSim to provide efficient accuracy using triangulation reducing detection errors.
- Data was collected and interpreted through MOT files where coordinates of specific joints were measured through code.

- **Data Preprocessing:**
  - Each trial’s time axis was scaled form 0 – 1 due to the offset timing from when recording stared.





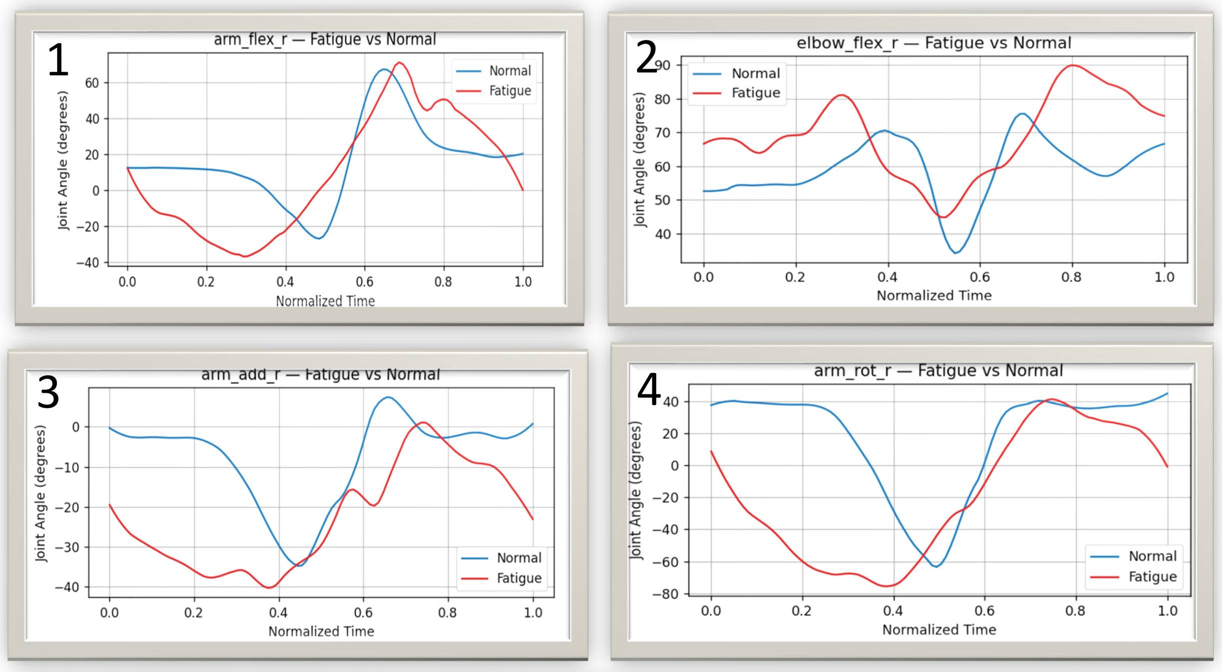
Euclidean Distance Formula: $d(x, y) = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2}$

- Averaging out the final data of 6 data sets for both categories (Fatigue vs Regular Forehand), only presented us with the “average swing” compared to self-standards.
- However, basing a conclusion off regular trials can be seen as **inaccurate**.
- By using dynamic time warping and the Euclidean norm, I was able to create a baseline by taking the average joint movement across time to form a representative "normal" curve.
- Euclidean Norm also gave the distance between the baseline trial vs “actual” trial where we would then make a “midpoint line” to receive the most accurate data set.

Results

Figure(s) 1- 4:

- Midline points were taken across the baseline trials to get an accurate representation of a “standard” line.
- L2 Distances Between Baseline and Fatigue:
  - arm_flex_r: 242.21
  - arm_add_r: 187.83
  - arm_rot_r: 517.74
  - elbow_flex_r: 156.00
- Among all joints analyzed, the greatest deviation from baseline was observed in the elbow flexion (elbow_flex_r), indicating a higher susceptibility to fatigue-induced variability or compensatory movement patterns during the forehand swing



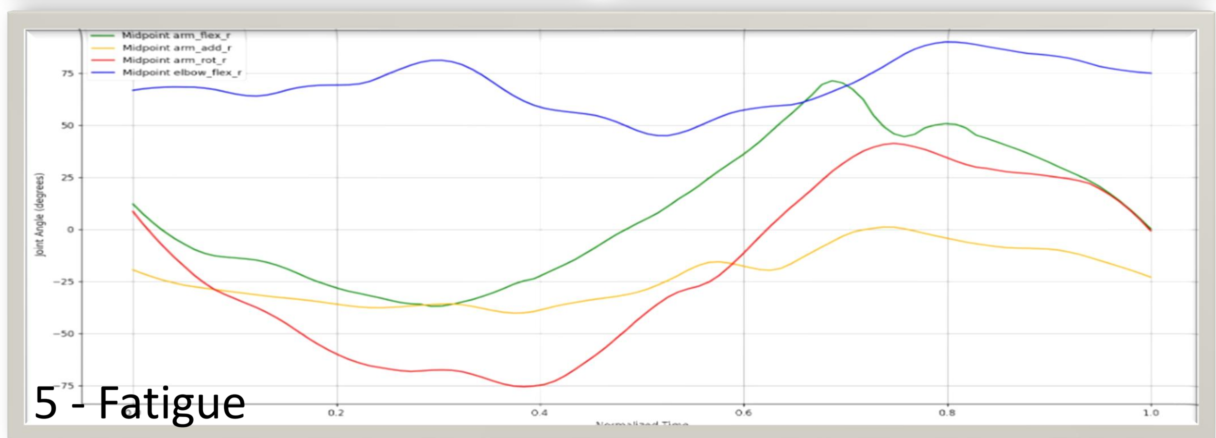
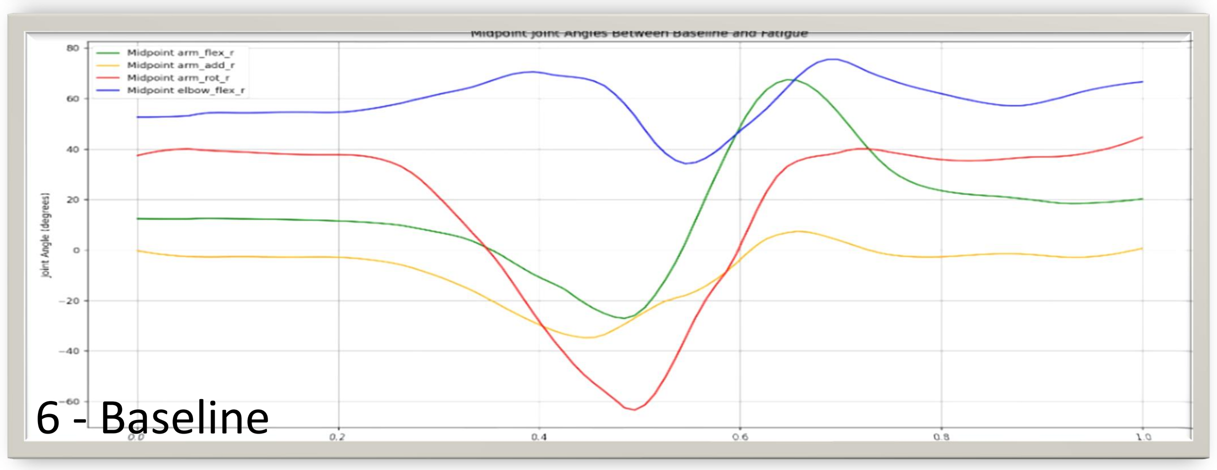
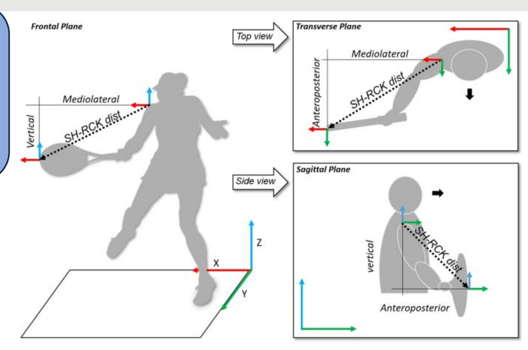


Figure 5 & 6:

- Relative to baseline, fatigue trials showed diminished peak joint angles and flattened angular trajectories, suggesting impaired neuromuscular responsiveness.



Discussion/Conclusion



Kinematic Changes Due to Fatigue:

- Flattened Joint Peaks:
  - Compared to baseline data, peak joint angles during the fatigue trials exhibited lower magnitude and reduced steepness (Fig 5, 6) suggesting a slower and less forceful contraction or extension during movement.
  - Fatigue movements that take longer to reach peak angular positions indicated diminished neuromuscular responsiveness.
  - Inconsistency in repetitive form can reflect a shift towards compensatory movement strategies which lead to joint misalignment and repetitive strain injuries.
- Timing and Joint Disruption
  - During forehand motion, peak elbow flexion typically happens around 40% - 60% of the movement cycle aligning the ball with the middle of the extended racquet to create optimal topspin.
  - In the fatigue trials, elbow contraction was delayed resulting in a weakened reactive control.
  - The observed joint instabilities under fatigue reinforce the need to optimize rest intervals under high performance. Rest periods longer than 30 may reduce neuromuscular decline and promote movement consistency.

References

- Buszard, T., Garofolini, A., Whiteside, D. *et al.* Children’s coordination of the “sweet spot” when striking a forehand is shaped by the equipment used. *Sci Rep* **10**, 21003 (2020). https://doi.org/10.1038/s41598-020-77627-5
- Alba-Jiménez C, Moreno-Doutres D, Peña J. Trends Assessing Neuromuscular Fatigue in Team Sports: A Narrative Review. Sports (Basel). 2022 Feb 28;10(3):33. doi: 10.3390/sports10030033. PMID: 35324642; PMCID: PMC8950744
- "Assessment of Tennis Elbow." Physiopedia, . 27 Oct 2024, 09:41 UTC. 27 Jul 2025, 09:35 https://www.physio-pedia.com/index.php?title=Assessment_of_Tennis_Elbow&oldid=362219
- OpenCap: 3D human movement dynamics from smartphone videos Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, Scott L. Delp
- Seth, Ajay, et al. “OpenSim: Simulating Musculoskeletal Dynamics and Neuromuscular Control to Study Human and Animal Movement.” *PLOS Computational Biology*, Public Library of Science, journals.plos.org/ploscompbiol/article?id=10.1371%2Fjournal.pcbi.1006223#ack. Accessed 27 July 2025.